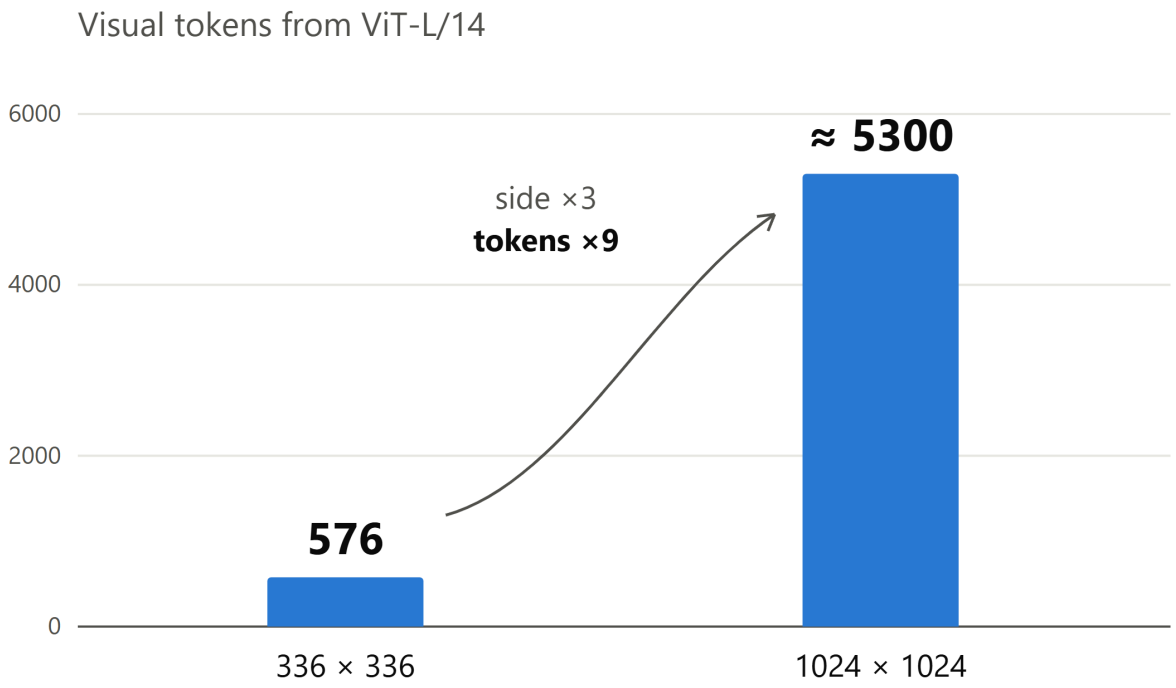
FastVLM

Redesigning the vision encoder for high resolution

CVPR 2025 · Apple

Higher resolution, slower VLM

TTFT = vision encoding time + LLM prefill time

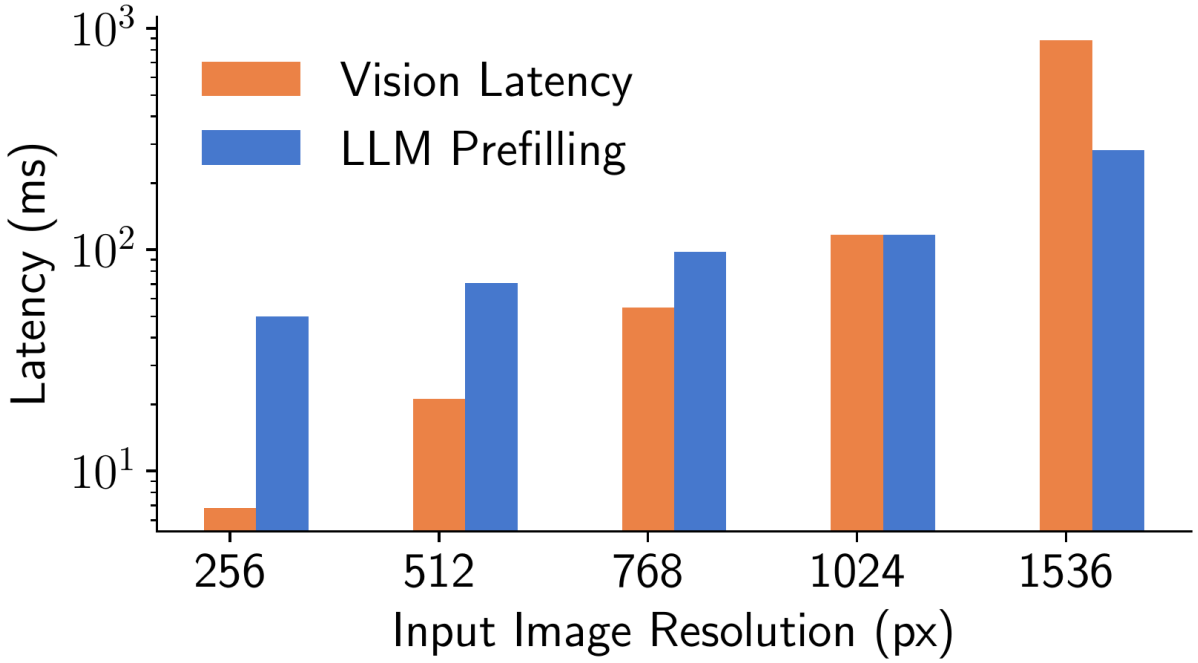


Token count grows with the **square** of resolution

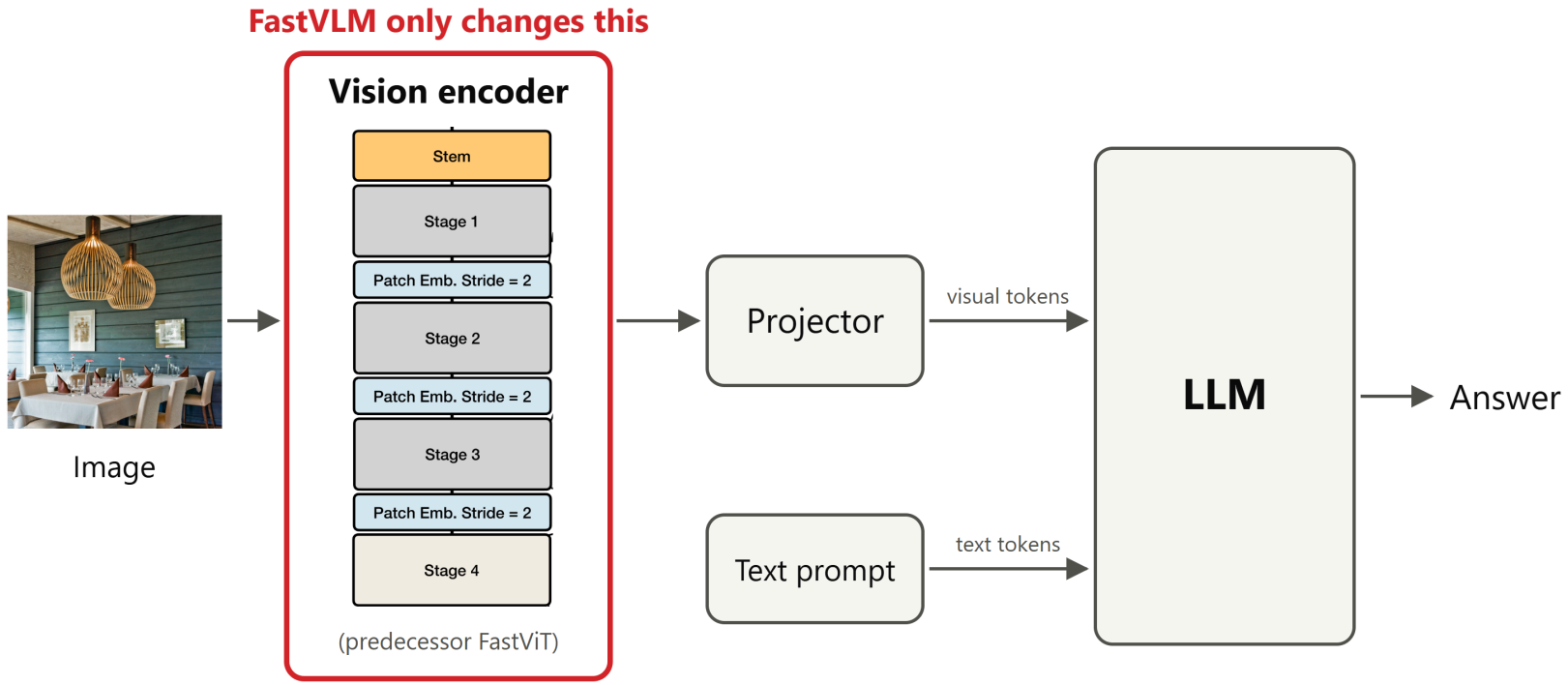
Cutting tokens is not enough

- At high resolution, **vision encoding itself** takes most of the time
- Token pruning, image tiling: patches applied outside the ViT

→ Can we use an encoder **built for high resolution**?

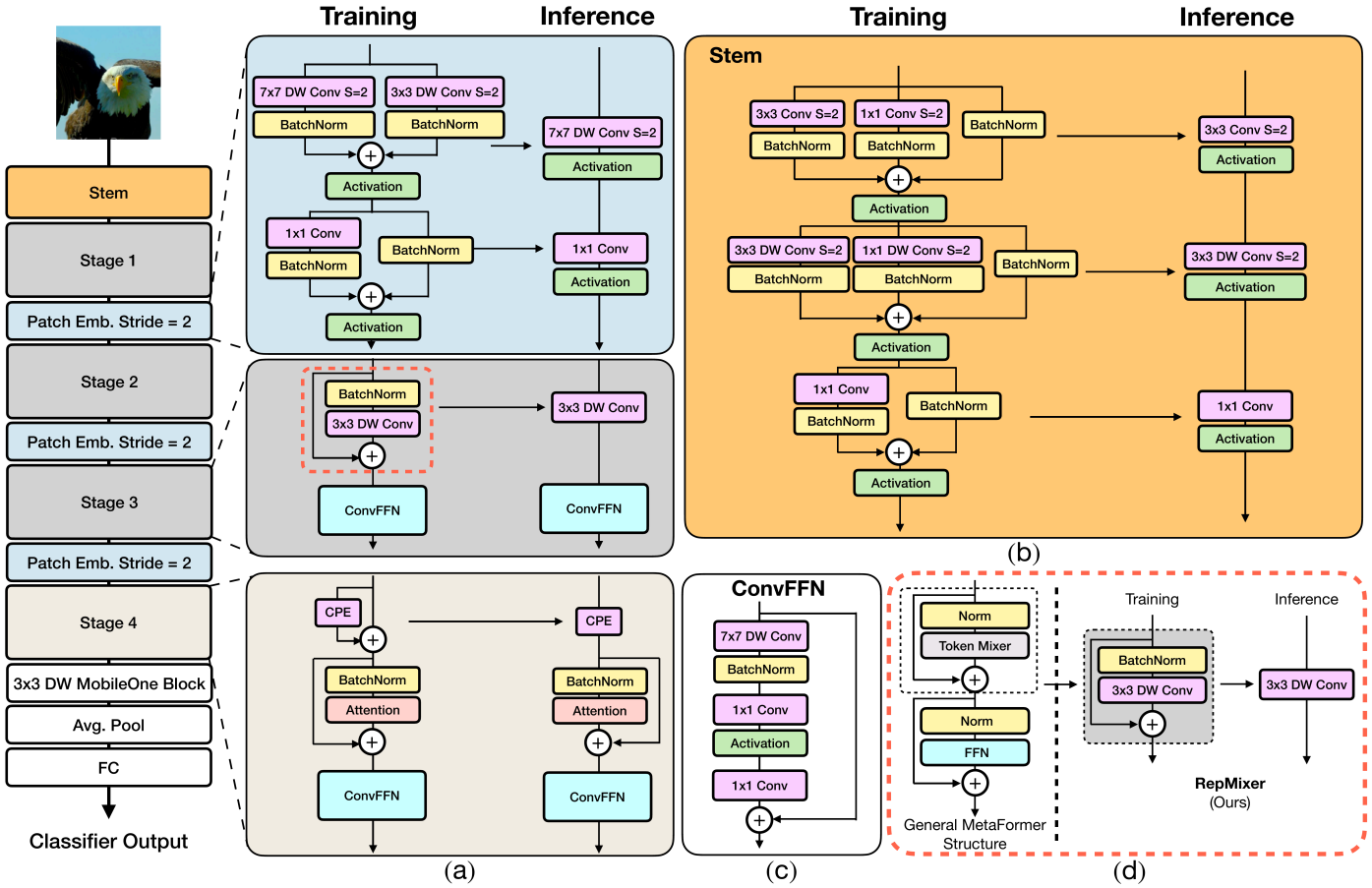


FastVLM swaps one box in the pipeline



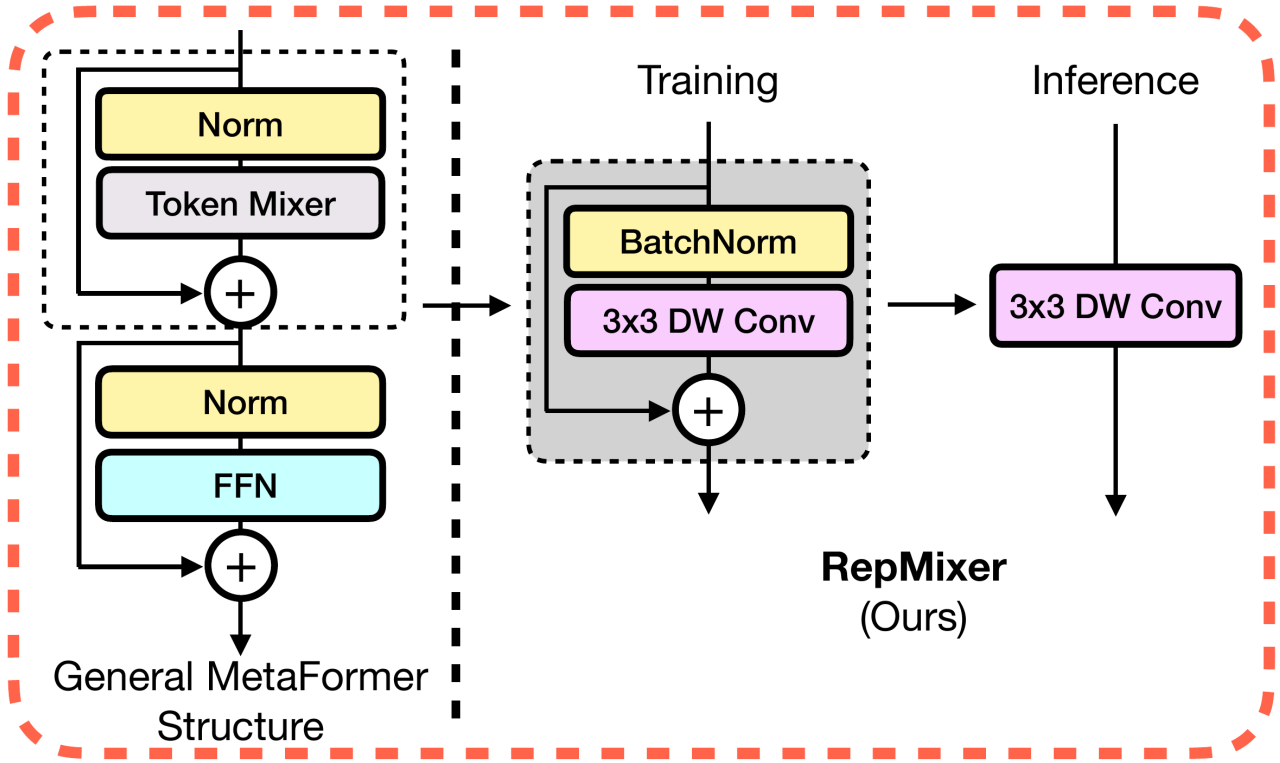
- Only the vision encoder changes; projector and LLM follow LLaVA
- Every stage from here on lives **inside** this box

Predecessor FastViT: the big picture



Left: the skeleton, a 4-stage pyramid | Right: the parts inside each block

The conv part is fine: one conv at inference



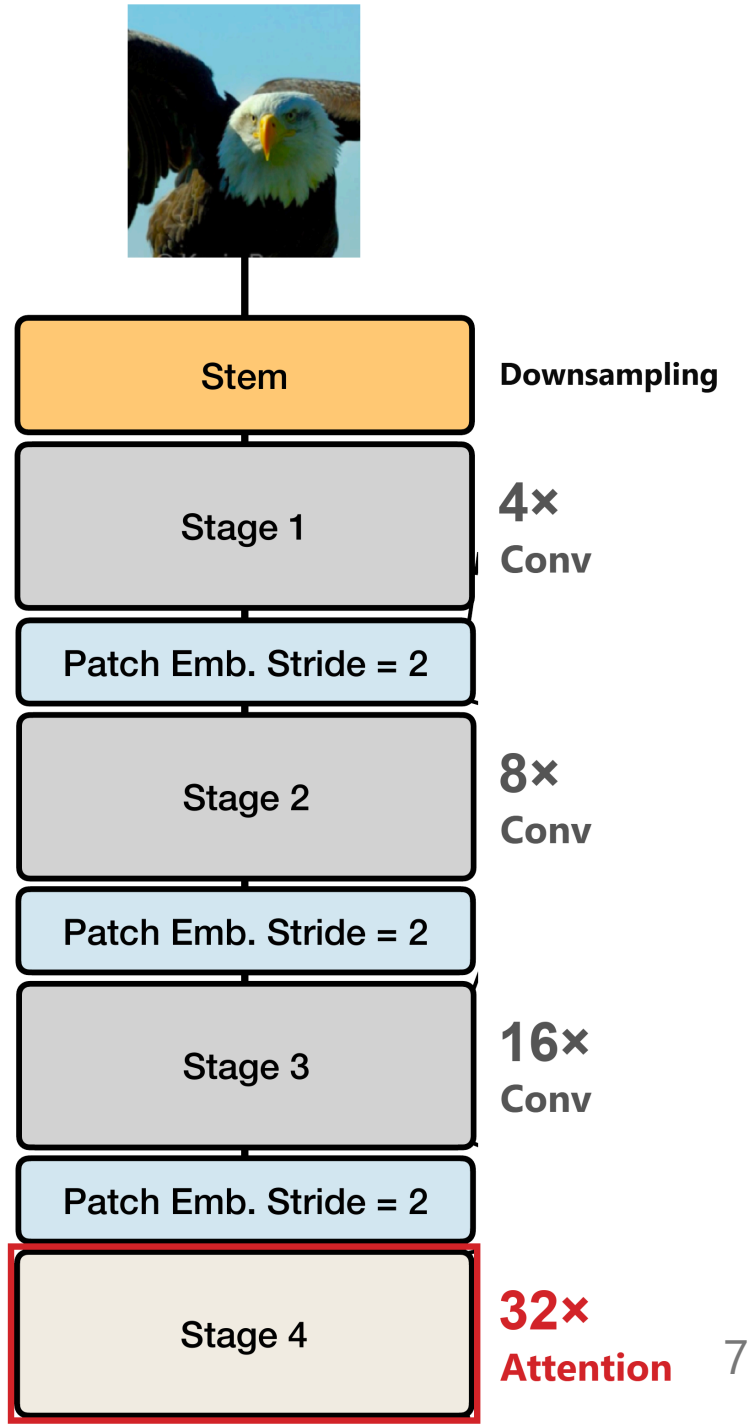
Training: several branches (easy to train) → Inference: merged into one conv (fast)

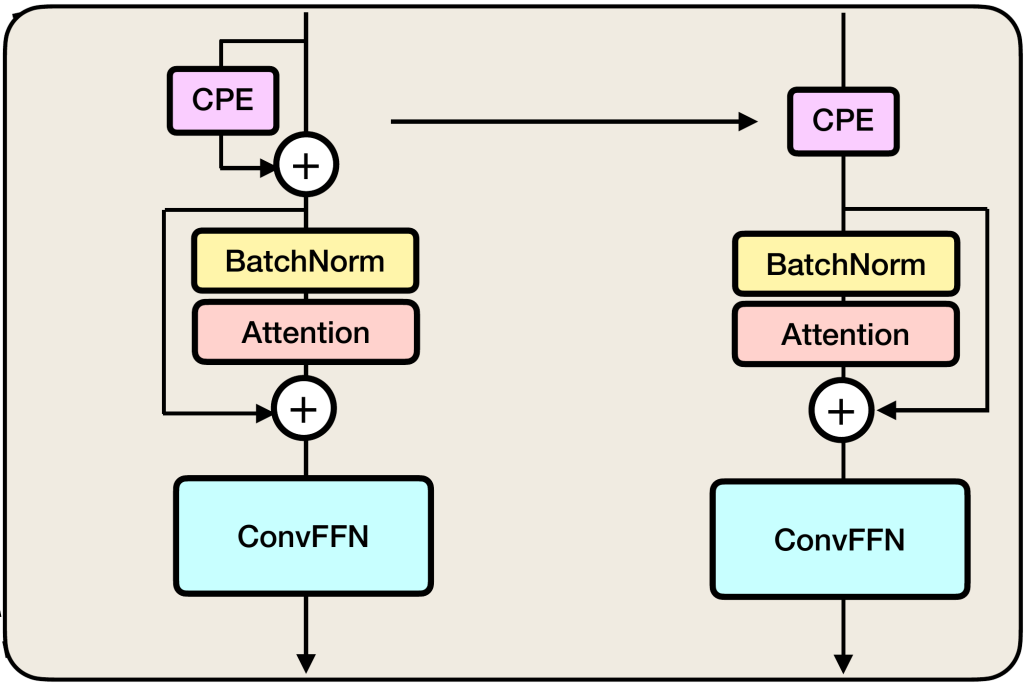
The bigger the image, the more it saves → **suits high resolution, keep as is**

Downsampled 32×: why is it still not enough?

- Each stage downsamples once; **every position left at the end is one token**
- 1024^2 image → 32×32 = **1024** tokens, still too many for the LLM

→ Can we **compress further**?



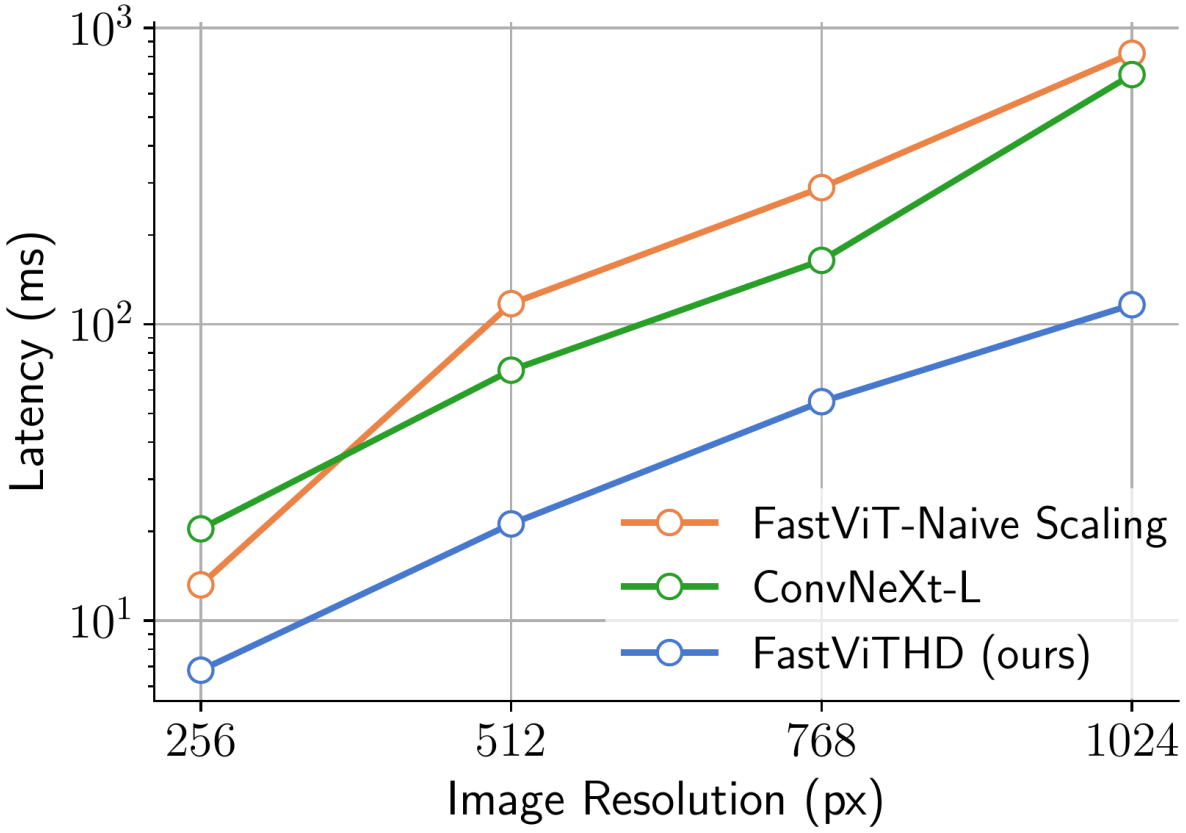


Self-attention comes too early

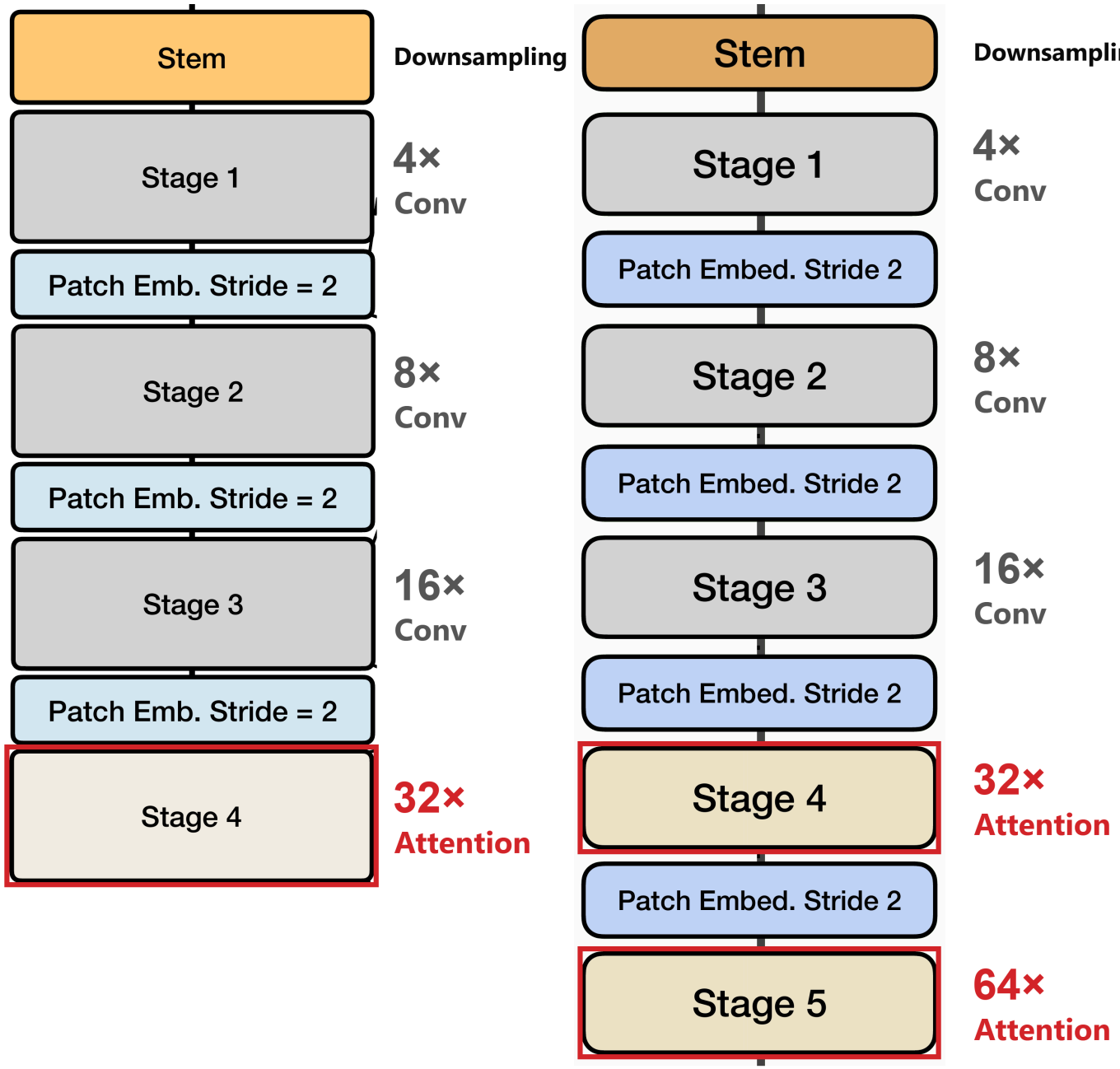
To scale a model up, the usual move is stacking self-attention in stages 3 and 4

Stage 3 (16× down)	Positions	Self-attention cost
224^2 image	196	cheap
1024^2 image	4096	about **400×** more

In practice: scaled the usual way, slower than pure conv



Orange: usual scaling | Green: pure-conv ConvNeXt-L | Blue: FastViTHD (next slide)



Compress one more level

- Add a 5th stage: 64× downsampling in total
- Move self-attention to the 32× and 64× stages

ViT	FastViT	FastViTHD
≈ 5300	1024	256

(tokens for a 1024^2 input)

Why didn't anyone do this earlier?

- The harder you compress, the more area one token covers → **small text gets lost**
- The view at the time: ViTamin recommends only 16× downsampling in total

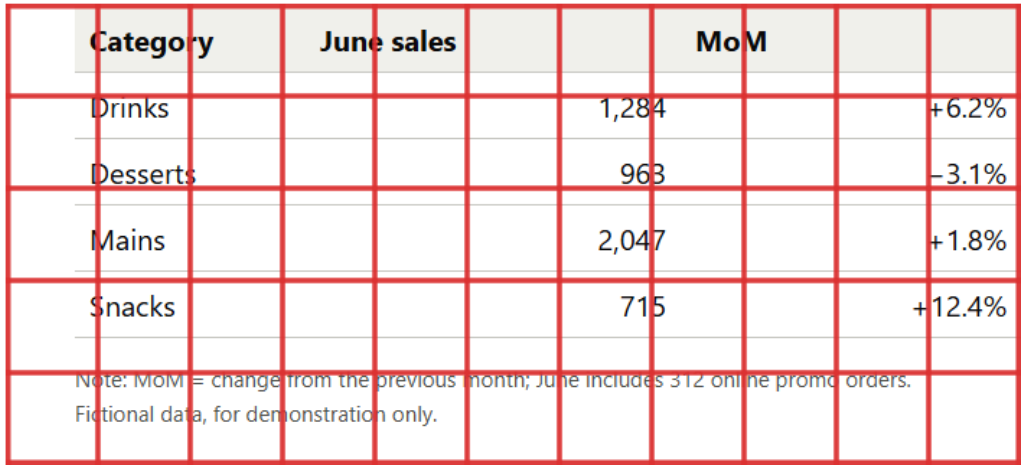
→ The real question: **how do we compress without losing information?**

32× downsampling: 1 cell = 32×32 px

Category	June sales	MoM
Drinks	1,284	+6.2%
Desserts	963	−3.1%
Mains	2,047	+1.8%
Snacks	715	+12.4%

Note: MoM = change from the previous month; June includes 312 online promo orders.
Fictional data, for demonstration only.

64× downsampling: 1 cell = 64×64 px

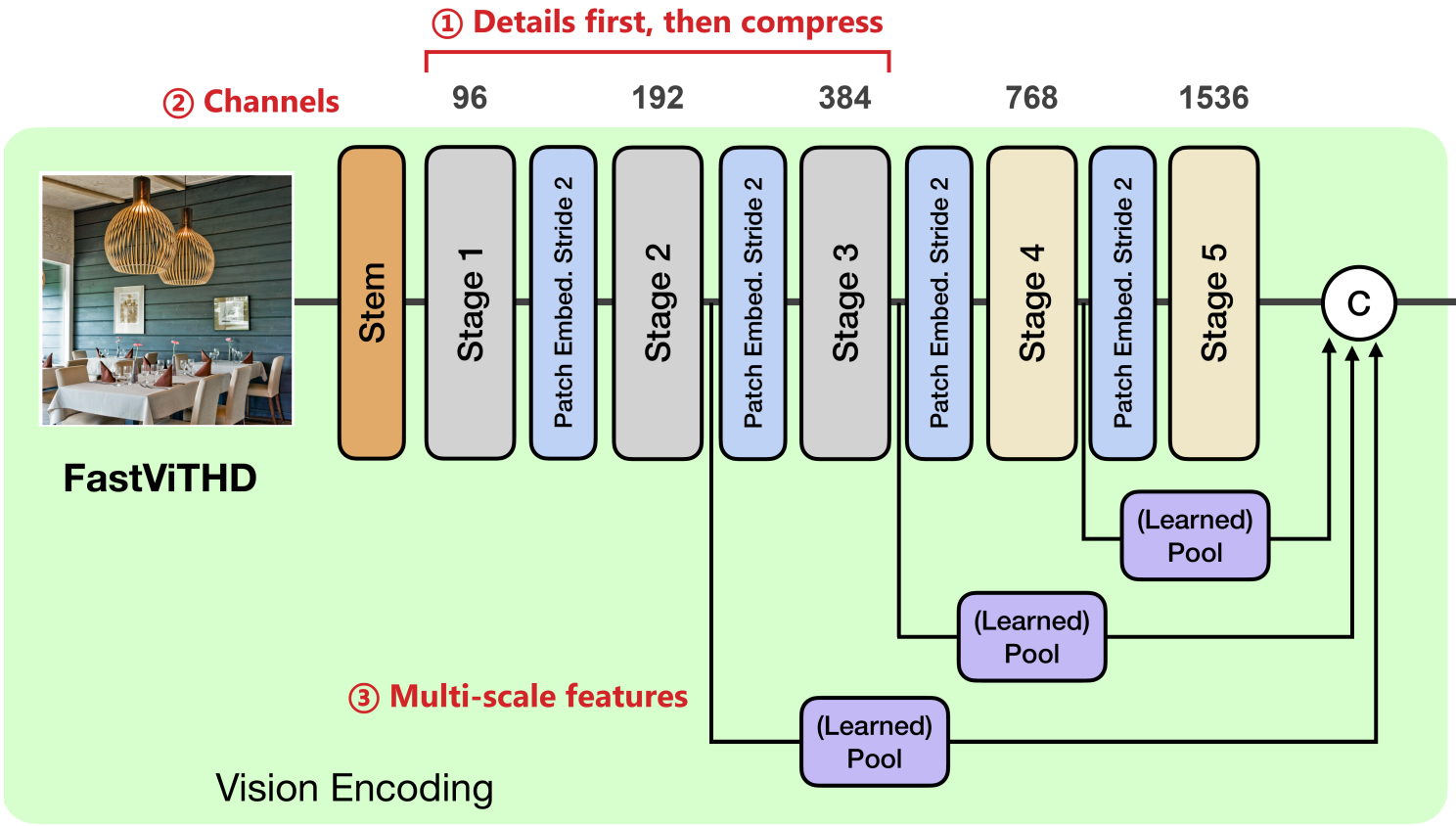


Compressing the right way

① Details first, then compress
Cheap convs (RepMixer) while the map is large; self-attention only once it is small

② Trade space for channels (next slide)

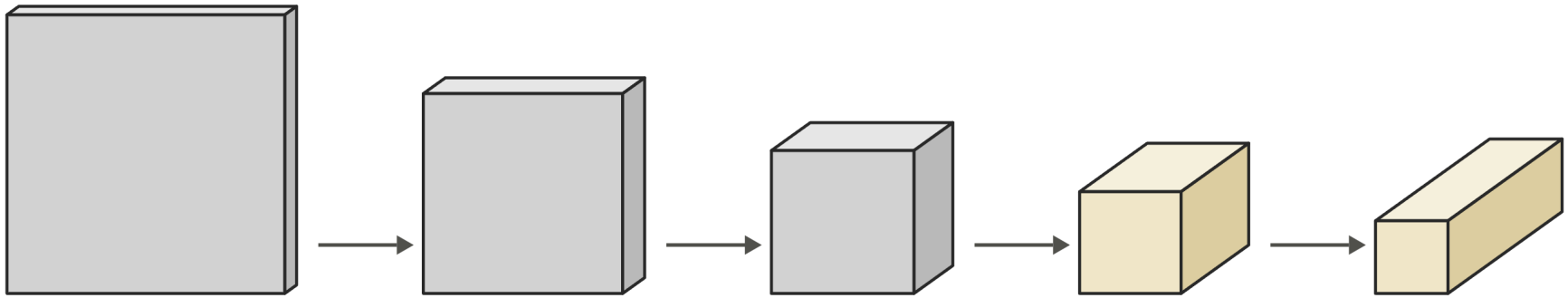
③ Multi-scale features (the slide after)



② Trading space for channels: what is a "channel"?

- Feature map = height × width × **channels**
- Each conv kernel outputs one channel; more kernels, more channels → **how many features** each position has

Front face = height × width (positions) · Depth = channels

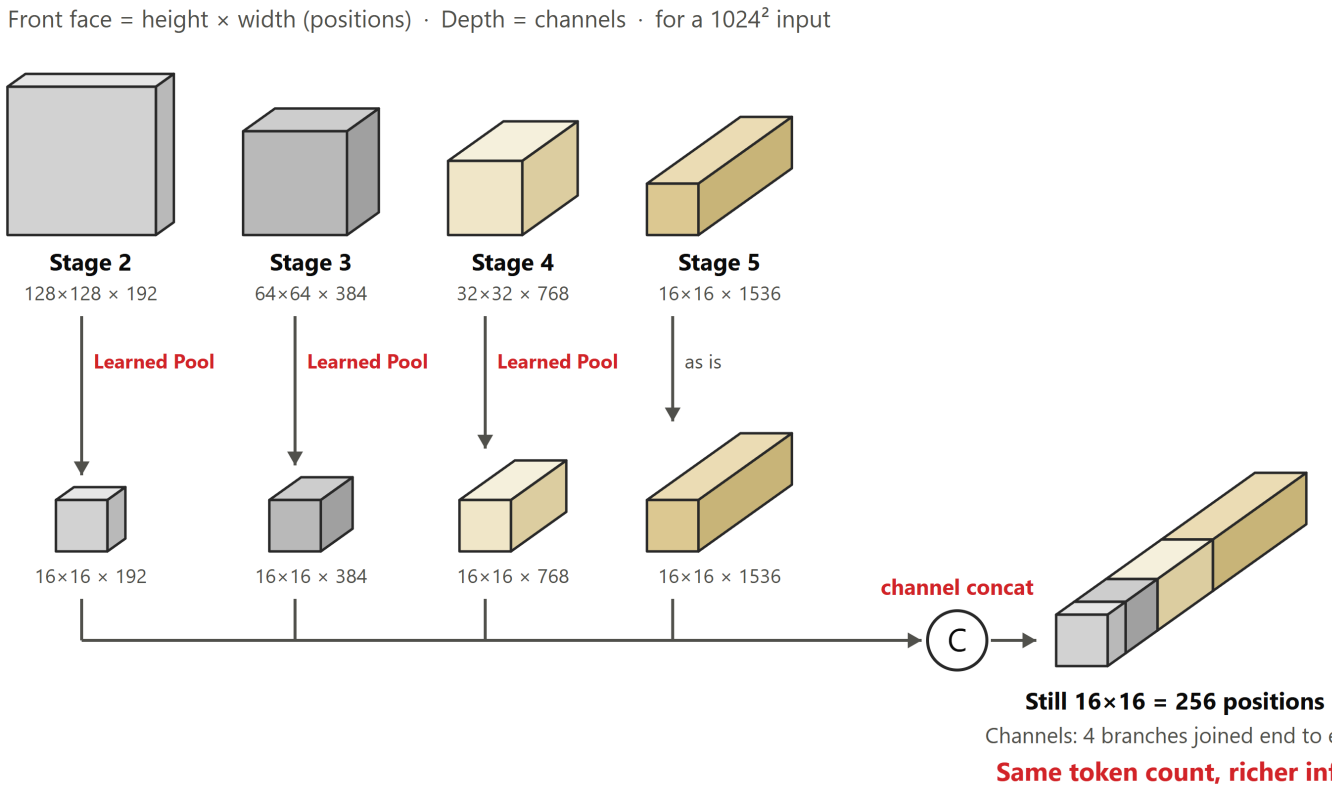


Each downsampling: height and width halve, channels double → each position holds more

③ Multi-scale: what is the C?

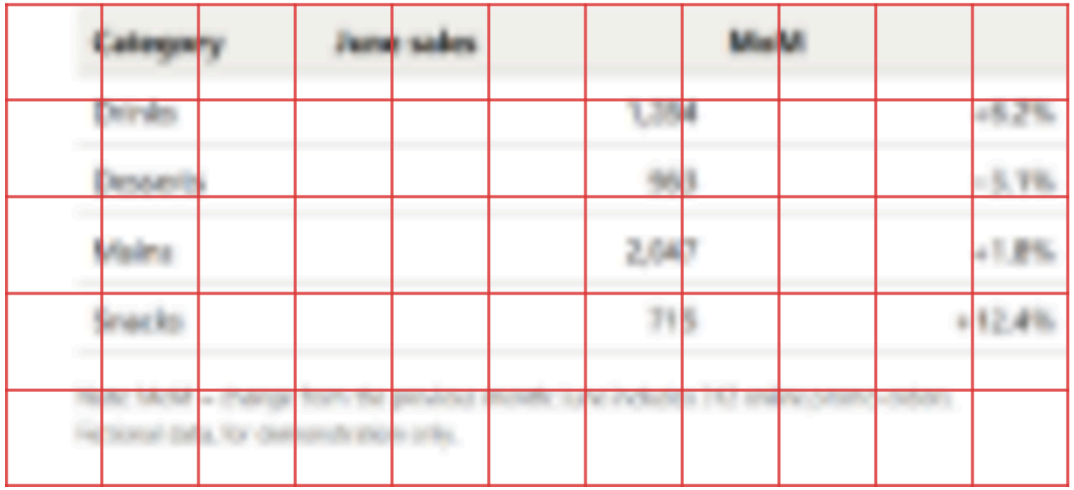
1. Take one branch each from stages 2, 3 and 4
2. **Learned Pool**: a strided depthwise conv shrinks each to stage 5's size (16×16)
3. **C**: concatenate the branches along the **channel** dimension

→ Still **256** tokens, but each token now carries both coarse and fine information



Compression buys resolution

ViT: 224×224, 256 tokens



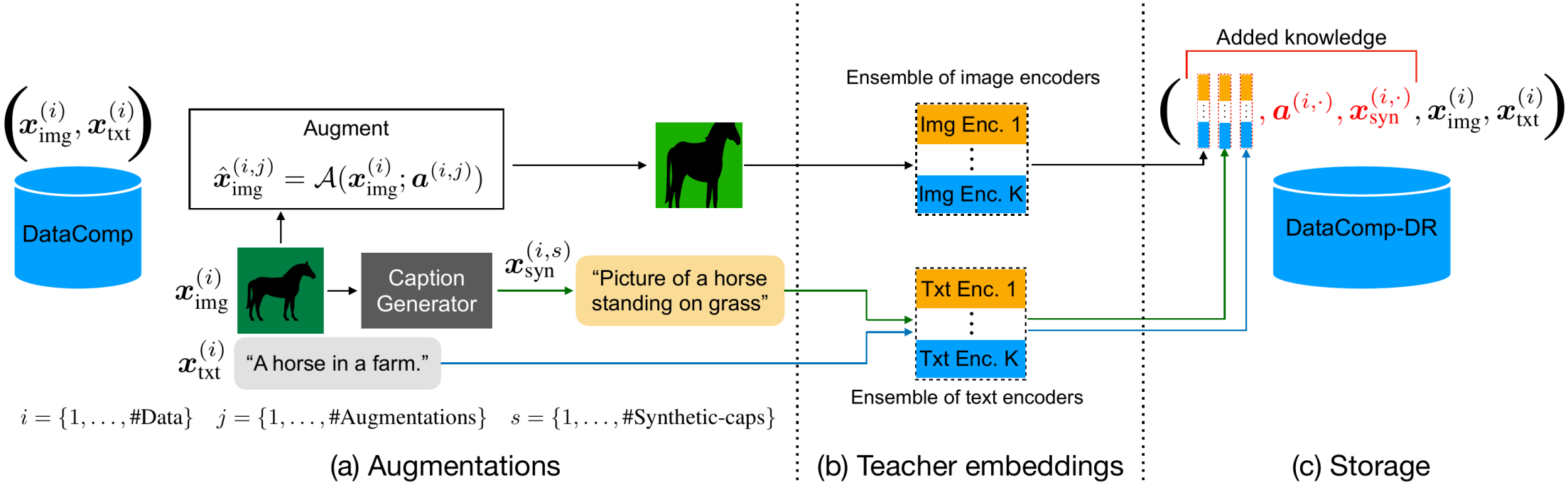
FastViTHD: 1024×1024, 256 tokens

Category	June sales	MoM
Drinks	1,284	+6.2%
Desserts	963	−3.1%
Mains	2,047	+1.8%
Snacks	715	+12.4%

Note: MoM = change from the previous month; June includes 312 online promo orders.
Fictional data, for demonstration only.

The same 256 tokens: for a VLM, **resolution is what matters**

Cost ①: harder compression needs stronger training



Each token must hold more information → pretraining has to be stronger

Taken from their MobileCLIP: prepare **better captions** and **large teacher models' outputs** for every image-text pair

Cost ②: resolution must match the LLM size

- A very small LLM **can't make use of** ultra-high-resolution images
- However well the vision side compresses, both ends still have to be tuned together

Redesign the encoder from the inside, instead of patching it from the outside